Hydration Risk Solver: Dynamic Physiological & Environmental Engine

This document provides a technical overview of how the **GetVari** Hydration Risk Solver dynamically calculates user dehydration risk. The engine synthesizes personal user parameters from the onboarding process with real-time biometric and weather telemetry to tailor safety scoring.

1. Core Mathematical Model
The baseline hydration risk score is computed using a weighted linear combination of five calculated physiological and environmental loads:

$$\text{Hydration Risk Score} = (w_{\text{heart}} \times \text{Heart Load}) + (w_{\text{activity}} \times \text{Activity Load}) + (w_{\text{temp}} \times \text{Temperature Load}) + (w_{\text{humidity}} \times \text{Humidity Load}) + (w_{\text{time}} \times \text{Time Load})$$

Base Loads Mapping (0 - 100)

Raw sensor readings are first mapped into standard **Base Loads** before applying user-
specific multipliers:

Metric	Range	Base Load
Heart Rate	<80 BPM | $80–100$ BPM | $100–120$ BPM | >120 BPM	10 | 40 | 70 | 90
Activity Index	<20 (Sedentary) | $20–50$ (Walk) | $50–80$ (Moderate) | ≥ 80 (Workout)	10 | 40 | 70 | 90
Temperature	$<22°$C | $22–28°$C | $28–32°$C | $>32°$C	10 | 30 | 60 | 90
Humidity	$<40\%$ | $40–60\%$ | $60–75\%$ | $>75\%$	20 | 40 | 70 | 90
Time Since Drink	<30 mins | $30–60$ mins | 1–2 hours | >2 hours	10 | 30 | 60 | 90

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2. Dynamic Adjustments (Onboarding Personalization)
The engine adjusts two parameters dynamically based on onboarding responses: **Multipliers** (m_i) applied to the base loads, and **Weights** (w_i) which are then normalized to sum to 1.0.

Coefficient Multipliers ($\boldsymbol{m_i}$)

- **Weight Exertion Scale**: If Weight >75 kg, activity coefficient multiplier increases: $m_{\text{activity}} = \min(1.4, \frac{\text{Weight}}{75})$.
- **Clinical Health Adjustments**:
  - *Diabetes*: $m_{\text{heart}} \times 1.15$, $m_{\text{time}} \times 1.10$
  - *Kidney Disease*: $m_{\text{time}} \times 1.25$
  - *Hypertension*: $m_{\text{heart}} \times 1.10$
  - *Chronic Heat-Stress*: $m_{\text{temp}} \times 1.20$
- **Climate Locations (Hot Cities)** (e.g., Mumbai, Chennai, Delhi, Jaipur): $m_{\text{temp}} \times 1.15$, $m_{\text{humidity}} \times 1.15$.

Weight Tweaks ($\boldsymbol{w_i}$)

- **Weight** >75 kg: $w_{\text{activity}} += 0.05$, $w_{\text{time}} -= 0.05$

- **Diabetes**: $w_{\text{heart}} += 0.05$, $w_{\text{time}} += 0.05$, $w_{\text{humidity}} -= 0.05$, $w_{\text{temp}} -= 0.05$
- **Kidney Disease**: $w_{\text{time}} += 0.10$, $w_{\text{humidity}} -= 0.05$, $w_{\text{temp}} -= 0.05$
- **Hypertension**: $w_{\text{heart}} += 0.05$, $w_{\text{activity}} -= 0.05$
- **Chronic Heat-Stress**: $w_{\text{temp}} += 0.05$, $w_{\text{time}} -= 0.05$
- **Hot Climates**: $w_{\text{temp}} += 0.05$, $w_{\text{humidity}} += 0.05$, $w_{\text{time}} -= 0.05$, $w_{\text{activity}} -= 0.05$

3. Concrete Use Case Scenarios

Use Case 1: Diabetic, Heavier Mass (85 kg) in Mumbai
- **Profile**: 85 kg, Mumbai (Hot/Humid), Diabetic, Moderate Exertion.
- **Telemetry Inputs**: HR = 90 BPM (Base=40), Activity Index = 60 (Base=70), Temp = 30°C (Base=60), Humidity = 65% (Base=70), Time since drink = 1.5 hours (Base=60).
- **Weights Applied (Normalized)**:
  - $w_{\text{heart}} = 0.35$, $w_{\text{activity}} = 0.25$, $w_{\text{temp}} = 0.20$, $w_{\text{humidity}} = 0.10$, $w_{\text{time}} = 0.10$
- **Multipliers & Final Loads**:
  - Heart Load $= 40 \times 1.15$ (Diabetic) $= 46.0$
  - Activity Load $= 70 \times 1.133$ (85kg) $= 79.31$
  - Temperature Load $= 60 \times 1.15$ (Mumbai) $= 69.0$
  - Humidity Load $= 70 \times 1.15$ (Mumbai) $= 80.5$
  - Time Load $= 60 \times 1.10$ (Diabetic) $= 66.0$
- **Calculation**: Risk Score $= (0.35 \times 46) + (0.25 \times 79.31) + (0.20 \times 69) + (0.10 \times 80.5) + (0.10 \times 66) = \mathbf{64.38}$
- **Result**: **64 (High Risk)**. (Without adjustments, the risk would be 57.0, showing how the engine proactively alerts this high-risk user).

Use Case 2: Healthy, Standard Weight (70 kg) in London
- **Profile**: 70 kg, London (Temperate/Cold), No Medical Conditions.
- **Telemetry Inputs**: HR = 90 BPM (Base=40), Activity Index = 60 (Base=70), Temp = 18°C (Base=10), Humidity = 50% (Base=40), Time since drink = 1.5 hours (Base=60).
- **Weights Applied (Normalized)**:
  - $w_{\text{heart}} = 0.30$, $w_{\text{activity}} = 0.25$, $w_{\text{temp}} = 0.20$, $w_{\text{humidity}} = 0.10$, $w_{\text{time}} = 0.15$
- **Multipliers & Final Loads**: All multipliers are default (1.0).
  - Loads: Heart $= 40$, Activity $= 70$, Temp $= 10$, Humidity $= 40$, Time $= 60$.
- **Calculation**: Risk Score $= (0.30 \times 40) + (0.25 \times 70) + (0.20 \times 10) + (0.10 \times 40) + (0.15 \times 60) = \mathbf{44.5}$
- **Result**: **45 (Mild Risk)**.

Use Case 3: Renal Patient (65 kg) in Bangalore (High Altitude / Dry)
- **Profile**: 65 kg, Bangalore (Moderate Climate), Kidney Disease.

- **Telemetry Inputs**: HR = 75 BPM (Base=10), Activity Index = 15 (Base=10), Temp = 25°C (Base=30), Humidity = 45% (Base=40), Time since drink = 2.5 hours (Base=90).
- **Weights Applied (Normalized)**:
  - $w_{\text{heart}} = 0.30$, $w_{\text{activity}} = 0.25$, $w_{\text{temp}} = 0.15$, $w_{\text{humidity}} = 0.05$, $w_{\text{time}} = 0.25$
- **Multipliers & Final Loads**:
  - Time Load $= 90 \times 1.25$ (Kidney Disease) $= 112.5 \rightarrow$ clamped to 100
  - Loads: Heart $= 10$, Activity $= 10$, Temp $= 30$, Humidity $= 40$, Time $= 100$.
- **Calculation**: Risk Score $= (0.30 \times 10) + (0.25 \times 10) + (0.15 \times 30) + (0.05 \times 40) + (0.25 \times 100) = \mathbf{37.0}$
- **Result**: **37 (Mild Risk)**. (Even though the patient is resting with low heart rate/activity, the prolonged time since drink triggers a higher risk score due to renal clearance requirements).

4. Dynamic Recovery & Prediction Models

When a user logs a fluid intake, the engine predicts the recovery results dynamically:
Recovery Window (Fluid Absorption Timeline)

Estimated time required for full fluid absorption is scaled by physiological constraints:

- **Base Timeline**: ≤ 200 ml $\rightarrow 25$ mins, ≤ 500 ml $\rightarrow 75$ mins, > 500 ml $\rightarrow 100$ mins.
- **Physiological Modifiers**:
  - *Weight*: Multiplied by $\min(1.3, \frac{\text{Weight}}{75})$ (larger circulatory and distribution volume requires longer absolute travel times).
  - *Diabetes*: Multiplied by $1.25 \times$ (slowed gastric emptying / hyperosmotic blood shift).
  - *Kidney Disease*: Multiplied by $1.40 \times$ (clearance limits to protect renal tubules).
  - *Hypertension*: Multiplied by $1.10 \times$.
  - *Hot Climates*: Multiplied by $0.90 \times$ (increased skin vasodilation draws fluid quicker to surface capillaries).

Risk Mitigation Forecast (Post-Intake Risk Prediction)
The predicted drop in risk score is scaled by the user's mass and absorption efficiency: Risk Reduction $=$ Fluid (ml) $\times 0.08 \times \left(\frac{70}{\text{Weight (kg)}}\right) \times$ Absorption Efficiency

- *Absorption Efficiency*: 1.0 (standard), 0.90 (Diabetes), 0.85 (Kidney Disease).
- *Meaning*: A larger body has a larger total body water (TBW) pool. Consequently, a fixed 250ml drink reduces overall risk index less in an 85kg person compared to a 50kg person.

5. Concrete Use Case Scenarios

Use Case 1: Diabetic, Heavier Mass (85 kg) in Mumbai

- **Profile**: 85 kg, Mumbai (Hot/Humid), Diabetic.
- **Intake Event**: User drinks 250 ml of water.
- **Estimation Calculations**:
    - **Recovery Window Multiplier**: 1.133 (Weight) $\times$ 1.25 (Diabetes) $\times$ 0.90 (Mumbai) $= \mathbf{1.275}$
    - **Est. Absorption Time**: 75 mins (base) $\times$ $1.275 = 95.6$ mins $\rightarrow$ $\mathbf{1}$h $\mathbf{20}$m to $\mathbf{1}$h $\mathbf{50}$m.
    - **Risk Reduction**: 250 ml $\times$ 0.08 $\times$ $(\frac{70}{85})$ $\times$ 0.90 (Diabetes) $= \mathbf{14.82} \rightarrow$ $\mathbf{15}$ point reduction.

Use Case 2: Renal Patient (65 kg) in Bangalore

- **Profile**: 65 kg, Bangalore (Moderate Climate), Kidney Disease.
- **Intake Event**: User drinks 200 ml of water.
- **Estimation Calculations**:
    - **Recovery Window Multiplier**: 1.0 (Weight) $\times$ 1.40 (Renal) $= \mathbf{1.40}$
    - **Est. Absorption Time**: 25 mins (base) $\times$ $1.40 = 35$ mins $\rightarrow$ $\mathbf{25}$ to $\mathbf{45}$ mins.
    - **Risk Reduction**: 200 ml $\times$ 0.08 $\times$ $(\frac{70}{65})$ $\times$ 0.85 (Renal) $= \mathbf{14.64} \rightarrow$ $\mathbf{15}$ point reduction.